Department of Computer Science

Banasthali Vidyapith

Software Requirement Specifications (SRS) and Software Design Specifications (SDS)

[Project Name]

[Group id]

Guided By:

[Teacher Name]

Submitted By:

[Student Name with RollNo.]
[Class]

Chapter 1 SRS

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1. Introduction

The following subsections of the Software Requirements Specifications (SRS) document should provide an overview of the entire SRS. The thing to keep in mind as you write this document is that you are telling what the system must do – so that designers can ultimately build it. Do not use this document for design!!!

1.1 Purpose

Identify the purpose of this SRS and its intended audience. In this subsection, describe the purpose of the particular SRS and specify the intended audience for the SRS.

1.2 Scope

In this subsection:

(1) Identify the software product(s) to be produced by name
(2) Explain what the software product(s) will, and, if necessary, will not do
(3) Describe the application of the software being specified, including relevant benefits, objectives, and goals
(4) Be consistent with similar statements in higher-level specifications if they exist

This should be an executive-level summary. Do not enumerate the whole requirements list here.

1.3 Definitions, Acronyms, and Abbreviations.

Provide the definitions of all terms, acronyms, and abbreviations required to properly interpret the SRS. This information may be provided by reference to one or more appendices in the SRS or by reference to documents. This information may be provided by reference to an Appendix.

1.4 Overview

In this subsection:

(1) Describe what the rest of the SRS contains
(2) Explain how the SRS is organized

Don't rehash the table of contents here. Point people to the parts of the document they are most concerned with. Customers/potential users care about section 2, developers care about section 3.

2. The Overall Description

Describe the general factors that affect the product and its requirements. This section does not state specific requirements. Instead, it provides a background for those requirements, which are defined in section 3, and makes them easier to understand. In a sense, this section tells the requirements in plain English for the consumption of the customer. Section3 will contain a specification written for the developers.

2.1 Product Perspective

Put the product into perspective with other related products. If the product is independent and totally self-contained, it should be so stated here. If the SRS defines a product that is a component of a larger system, as frequently occurs, then this subsection relates the requirements of the larger system to functionality of the software and identifies interfaces between that system and the software. If you are building a real system, compare its similarity and differences to other systems in the marketplace. If you are doing a research-oriented project, what related research compares to the system you are planning to build.

A block diagram showing the major components of the larger system, interconnections, and external interfaces can be helpful. This is not a design or architecture picture. It is more to provide context, especially if your system will interact with external actors. The system you are building should be shown as a black box. Let the design document present the internals.

The following subsections describe how the software operates inside various constraints.

2.1.1 Product Function

Provide a summary of the major functions that the software will perform. Sometimes the function summary that is necessary for this part can be taken directly from the section of the higher-level specification (if one exists) that allocates particular functions to the software product.

For clarity:

(1) The functions should be organized in a way that makes the list of functions understandable to the customer or to anyone else reading the document for the first time.

(2) Textual or graphic methods can be used to show the different functions and their relationships. Such a diagram is not intended to show a design of a product but simply shows the logical relationships among variables.

AH, finally the real meat of section 2. This describes the functionality of the system in the language of the customer. What specifically does the system that will be designed have to do? Drawings are good, but remember this is a description of what the system needs to do, not how you are going to build it. (That comes in the design document).

2.1.2 Hardware Interfaces

Specify the logical characteristics of each interface between the software product and the hardware components of the system. This includes configuration characteristics. It also covers such matters as what devices are to be supported, how they are to be supported and protocols. This is not a description of hardware requirements in the sense that "This program must run on a Mac with 64M of RAM". This section is for detailing the actual hardware devices your application will interact with and control. For instance, if you are controlling X10 type home devices, what is the interface to those devices? Designers should be able to look at this and know what hardware they need to worry about in the design. Many business type applications will have no hardware interfaces. If none, just state "The system has no hardware interface requirements" If you just delete sections that are not applicable, then readers do not know if: a. this does not apply or b. you forgot to include the section in the first place.

2.1.3 Software Interfaces

Specify the use of other required software products and interfaces with other application systems. For each required software product, include:

(1) Name
(2) Mnemonic
(3) Specification number
(4) Version number
(5) Source

For each interface, provide:

(1) Discussion of the purpose of the interfacing software as related to this software product
(2) Definition of the interface in terms of message content and format

Here we document the APIs, versions of software that we do not have to write, but that our system has to use. For instance if your customer uses SQL Server 7 and you are required to use that, then you need to specify i.e.

2.1.4.1 Microsoft SQL Server 7. The system must use SQL Server as its database component. Communication with the DB is through ODBC connections. The system must provide SQL data table definitions to be provided to the company DBA for setup.

A key point to remember is that you do NOT want to specify software here that you think would be good to use. This is only for ***customer-specified systems*** *that you* ***have*** *to interact with. Choosing SQL Server 7 as a DB without a customer requirement is a Design choice, not a requirement. This is a subtle but important point to writing good requirements and not over-constraining the design.*

2.1.4 Communications Interfaces

Specify the various interfaces to communications such as local network protocols, etc. These are protocols you will need to directly interact with. If you happen to use web services transparently to your application then do not list it here. If you are using a custom protocol to communicate between systems, then document that protocol here so designers know what to design. If it is a standard protocol, you can reference an existing document or RFC.

2.2 User Characteristics

Describe those general characteristics of the intended users of the product including educational level, experience, and technical expertise. Do not state specific requirements but rather provide the reasons why certain specific requirements are later specified in section 3.

What is it about your potential user base that will impact the design? Their experience and comfort with technology will drive UI design. Other characteristics might actually influence internal design of the system.

2.3 Constraints

Provide a general description of any other items that will limit the developer's options. These can include:

(1) Regulatory policies
(2) Hardware limitations (for example, signal timing requirements)
(3) Interface to other applications
(4) Parallel operation
(5) Audit functions
(6) Control functions
(7) Higher-order language requirements
(8) *Signal handshake protocols (for example, XON-XOFF, ACK-NACK)*
(1) *Reliability requirements*
(10) Criticality of the application
(11) Safety and security considerations

This section captures non-functional requirements in the customers language. A more formal presentation of these will occur in section 3.

2.4 Technologies Used

3. Specific Requirements

This section contains all the software requirements at a level of detail sufficient to enable designers to design a system to satisfy those requirements, and testers to test that the system satisfies those requirements. Throughout this section, every stated requirement should be externally perceivable by users, operators, or other external systems. These requirements should include at a minimum a description of every input (stimulus) into the system, every output (response) from the system and all functions performed by the system in response to an input or in support of an output. The following principles apply:

(1) Specific requirements should be stated with all the characteristics of a good SRS
- *correct*
- *unambiguous*
- *complete*
- *consistent*

- *ranked for importance and/or stability*
- *verifiable*
- *modifiable*
- *traceable*

(2) Specific requirements should be cross-referenced to earlier documents that relate
(3) All requirements should be uniquely identifiable (usually via numbering like 3.1.2.3)
(4) Careful attention should be given to organizing the requirements to maximize readability (Several alternative organizations are given at end of document)

Before examining specific ways of organizing the requirements it is helpful to understand the various items that comprise requirements as described in the following subclasses. This section reiterates section 2, but is for developers not the customer. The customer buys in with section 2, the designers use section 3 to design and build the actual application.

Remember this is not design. Do not require specific software packages, etc unless the customer specifically requires them. Avoid over-constraining your design. Use proper terminology:

The system shall… A required, must have feature

The system should… A desired feature, but may be deferred till later

The system may… An optional, nice-to-have feature that may never make it to implementation.

Each requirement should be uniquely identified for traceability. Usually, they are numbered 3.1, 3.1.1, 3.1.2.1 etc. Each requirement should also be testable. Avoid imprecise statements like, "The system shall be easy to use" Well no kidding, what does that mean? Avoid "motherhood and apple pie" type statements, "The system shall be developed using good software engineering practice"

Avoid examples, This is a specification, a designer should be able to read this spec and build the system without bothering the customer again. Don't say things like, "The system shall accept configuration information such as name and address." The designer doesn't know if that is the only two data elements or if there are 200. List every piece of information that is required so the designers can build the right UI and data tables.

3.1 Functional

Functional requirements define the fundamental actions that must take place in the software in accepting and processing the inputs and in processing and generating the outputs. These are generally listed as "shall" statements starting with "The system shall…

These include:

- *Validity checks on the inputs*
- *Exact sequence of operations*
- *Responses to abnormal situation, including*
    - *Overflow*
    - *Communication facilities*
    - *Error handling and recovery*

- *Effect of parameters*
- *Relationship of outputs to inputs, including*
    - *Input/output sequences*
    - *Formulas for input to output conversion*

It may be appropriate to partition the functional requirements into sub-functions or sub-processes. This does not imply that the software design will also be partitioned that way.

3.2 Non functional Requirements(Software System Attributes)

3.2.1 AVAILABILITY

The availability of this web-site is up to the Internet connection of the client. Since this is client-server related web-site shall be attainable all the time. User should have an account to enter the system; if user does not have an account then user can only see the information which will be displayed on the homepage of the web-site.

3.2.2 SECURITY

The authorization mechanism of the system will block the unwanted attempts to the server and also let the system decide on which privileges may the user have. The system has different types of users so there are different levels of authorization.

3.2.3 RELIABILITY

A backup file is maintained so that in case of system crash, the data will not be affected.

3.2.4 PORTABILITY

The system is developed using ASP.Net which provides a framework for developing web based applications.

3.2.5 MAINTAINABILITY

This website will follow the modular structure so it will be easy to maintain.

Chapter 2 SDS

TABLE OF CONTENTS

1. Introduction

The following subsections of the Software Design Specifications (SDS) document should provide an overview of the entire SDS. The thing to keep in mind as you write this document is that you are telling what the system must do – so that designers can ultimately build it. Do not use this document for design!!!

1.1 Purpose

Identify the purpose of this SDS and its intended audience. In this subsection, describe the purpose of the particular SDS and specify the intended audience for the SDS.

1.2 Scope

In this subsection:

(5) Identify the software product(s) to be produced by name
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1.4 Overview

In this subsection:

(3) Describe what the rest of the SDS contains
(4) Explain how the SDS is organized

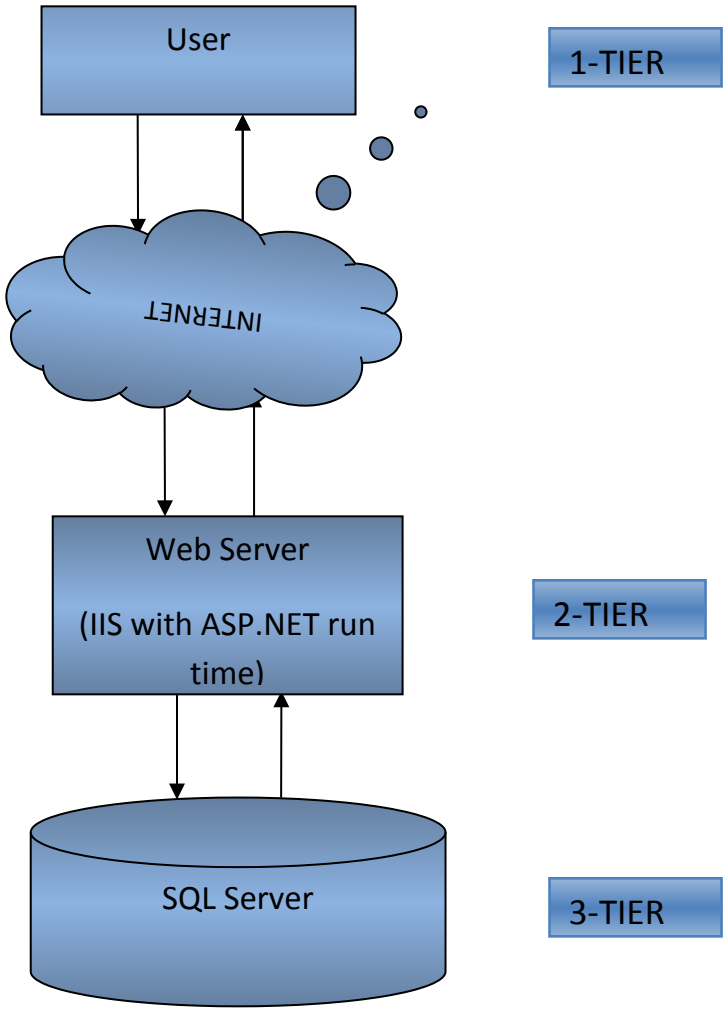
2. System Architectural Design

NOTE: This section is the main focus of the high-level conceptual design. The reader should come away with a good view of exactly how your solution is to be organized.

2.1 High-level Design Overview

Introduce the various components and systems at a high conceptual level. A

Pictorial representation of the system architecture is presented.



2.2) Detailed Description of Components

NOTE: This section is the main focus of the technical design portion of the SDS, the detailed design. This section will provide most of the basis for implementing the product.

Description for Component n

A detailed description of each software component contained within the architecture is presented. A description of each major software function along with:

Examples:-

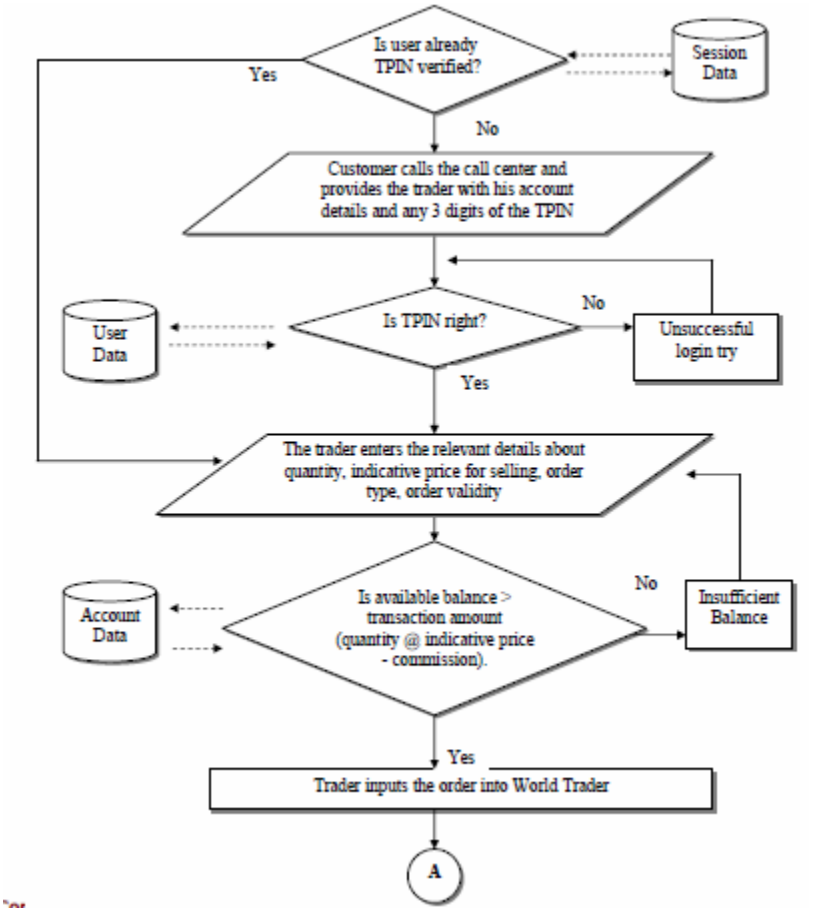
Data flow diagram (DFD) and Structure chart, Use Case diagram, Entity relationship diagram, Class diagram, Activity diagram and Sequence chart.

Component and interface diagram

A detailed description of the input and output interfaces for the component is presented.

Component and processing detail

A detailed algorithmic description for each component is presented. A diagram showing the flow of information through the function and the transformation it undergoes is presented. For Example –



3 Structure and relationships

This subsection will make clear the interrelationships and dependencies between modules/layers/components. Structure charts can be useful here. Another good idea is to use a simple finite state machine that demonstrates the operation of the product you are designing. There should always be explanatory text to help the reader understand any charts.

3.1) Data design

A description of all data structures including internal, global, and temporary data structures.

Temporary data structure

Files created for interim use are described.

StaffMember Table

Purpose: This table is used for storing all details related to a staff.

Field	Type	Description	Constraints
StaffMemberId	Varchar(9)	Id representing a Staff	
StaffName	Varchar(90)	Name of Employee	Not null
StaffType	Varchar(1)	Admin Backoffice Trader	
Profile	Varchar(100)	A string of menu id's for the front end, stating which functions a user can access.	
StaffStatus	Varchar(9)	This field is for Maker-Checker concept. It can be any one of the following values – AddPending','ModifyPending','Ready'	

Primary Key: Staff Member Id

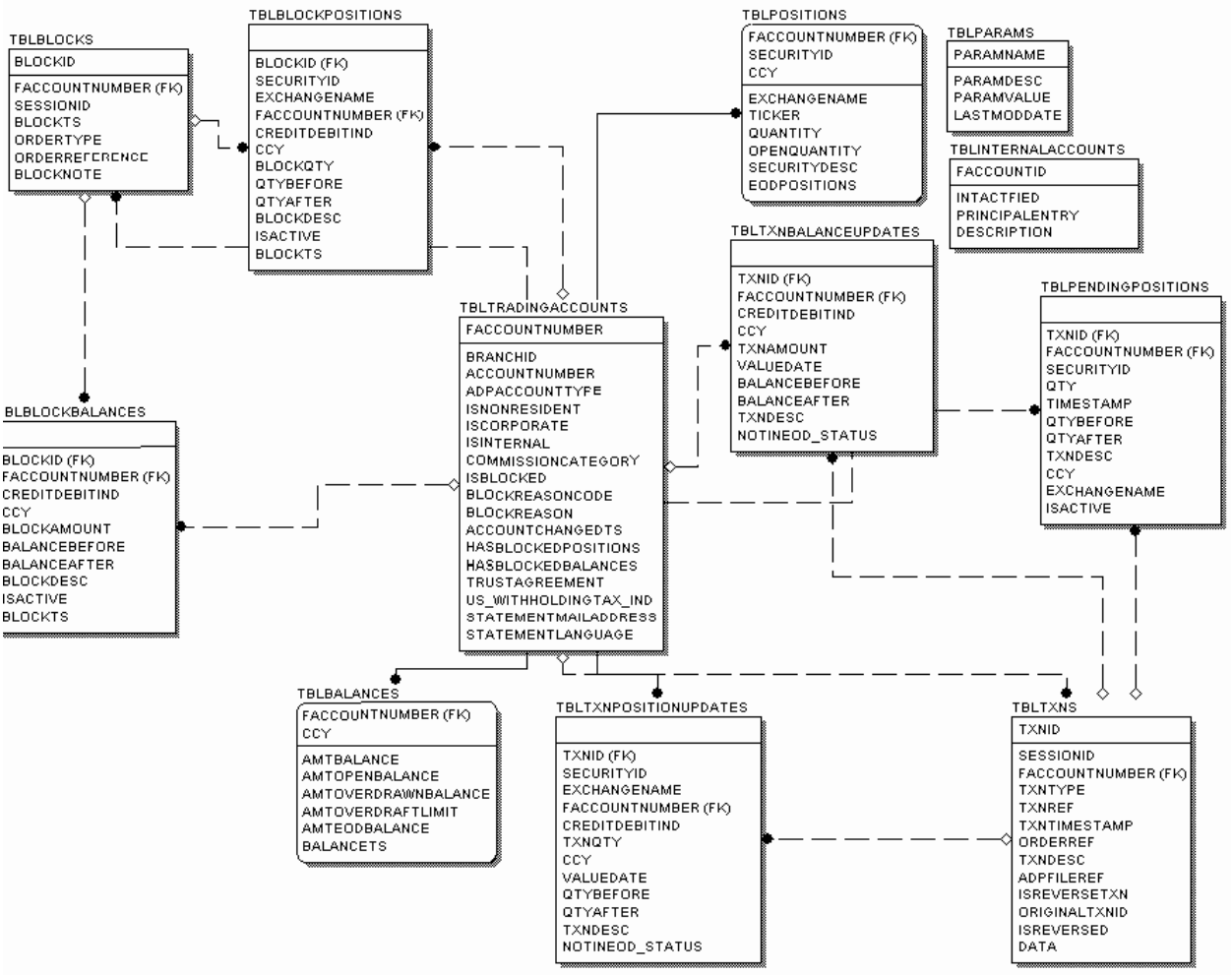
Foreign Key: staffStatus references 'paramCode' in systemParamTbl

Methods using this table

- **AddStaffMember**
- **ModifyStaffMember**
- **AddPasswordLogon**
- **StaffMemberList**

3.2 Database description

Database(s) created as part of the application is (are) described. E-R Model should clearly specify the data entities and relationship between them. After normalization, each table must be described with purpose, methods using it, all its fields description, their data type, and integrity constraints.


TBLBLOCKS
BLOCKID
FACCOUNTNUMBER (FK)
SESSIONID
BLOCKTS
ORDERTYPE
ORDERREFERENCE
BLOCKNOTE
TBLBLOCKPOSITIONS
BLOCKID (FK)
SECURITYID
EXCHANGENAME
FACCOUNTNUMBER (FK)
CREDITDEBITIND
CCY
BLOCKQTY
QTYBEFORE
QTYAFTER
BLOCKDESC
ISACTIVE
BLOCKTS
TBLPOSITIONS
FACCOUNTNUMBER (FK)
SECURITYID
CCY
EXCHANGENAME
TICKER
QUANTITY
OPENQUANTITY
SECURITYDESC
EODPOSITIONS
TBLPARAMS
PARAMNAME
PARAMDESC
PARAMVALUE
LASTMODDATE
TBLINTERNALACCOUNTS
FACCOUNTID
INTACTFIED
PRINCIPALENTRY
DESCRIPTION
TBLTXNBALANCEUPDATES
TXNID (FK)
FACCOUNTNUMBER (FK)
CREDITDEBITIND
CCY
TXNAMOUNT
VALUEDATE
BALANCEBEFORE
BALANCEAFTER
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TBLPENDINGPOSITIONS
TXNID (FK)
FACCOUNTNUMBER (FK)
SECURITYID
QTY
TIMESTAMP
QTYBEFORE
QTYAFTER
TXNDESC
CCY
EXCHANGENAME
ISACTIVE
TBLTRADINGACCOUNTS
FACCOUNTNUMBER
BRANCHID
ACCOUNTNUMBER
ADPACCOUNTTYPE
ISNONRESIDENT
ISCORPORATE
ISINTERNAL
COMMISSIONCATEGORY
ISBLOCKED
BLOCKREASONCODE
BLOCKREASON
ACCOUNTCHANGEDTS
HASBLOCKEDPOSITIONS
HASBLOCKEDBALANCES
TRUSTAGREEMENT
US_WITHHOLDINGTAX_IND
STATEMENTMAILADDRESS
STATEMENTLANGUAGE
BLBLOCKBALANCES
BLOCKID (FK)
FACCOUNTNUMBER (FK)
CREDITDEBITIND
CCY
BLOCKAMOUNT
BALANCEBEFORE
BALANCEAFTER
BLOCKDESC
ISACTIVE
BLOCKTS
TBLBALANCES
FACCOUNTNUMBER (FK)
CCY
AMTBALANCE
AMTOPENBALANCE
AMTOVERDRAWNBALANCE
AMTOVERDRAFTLIMIT
AMTEODBALANCE
BALANCETS
TBLTXNPOSITIONUPDATES
TXNID (FK)
SECURITYID
EXCHANGENAME
FACCOUNTNUMBER (FK)
CREDITDEBITIND
TXNQTY
CCY
VALUEDATE
QTYBEFORE
QTYAFTER
TXNDESC
NOTINEOD_STATUS
TBLTXNS
TXNID
SESSIONID
FACCOUNTNUMBER (FK)
TXNTYPE
TXNREF
TXNTIMESTAMP
ORDERREF
TXNDESC
ADPFILEREF
ISREVERSETXN
ORIGINALTXNID
ISREVERSED
DATA

4) User Interface Design

A description of the user interface design of the software is presented.

4.1 Description of the user interface

A detailed description of user interface including screen images or prototype is presented.

4.2 Screen images

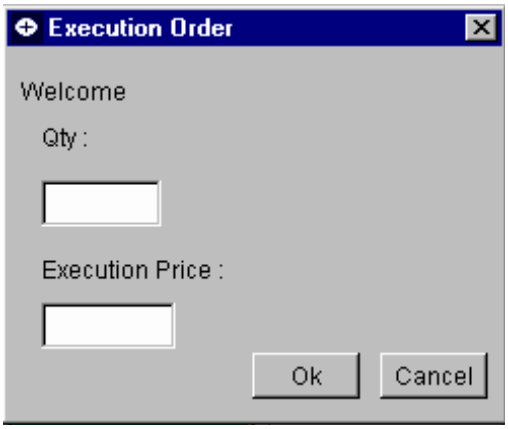
Representation of the interface form from the user's point of view.

4.3 Objects and actions

All screen objects and actions are identified. Describe each object's purpose, data type, properties and responses for different events.

For Example –

The Execution Order Function



5. TYPES OF TESTS (With Implementation)

Unit Tests

Unit tests are most commonly done by developers on their own machines or on a common server that is very volatile. It is not necessary that the unit test machines be the same platform and operating system as the target deployment environment, but the movement from the unit test environment to other testing environments should not require material code changes by developers. A plan for one machine per developer plus one small server should be included in the overall system architecture.

System Tests

The system test environment allows multiple modules to be connected together and executed as in a typical use-case scenario. The choice as to whether this is done on a separate machine from unit testing is up to the implementation and test team. If the target deployment environment is different from the unit test environment, the system test environment should contain a machine that matches the target environment. Although the system test machine need not match the size of the deployment box, it should have the same platform and operating system. A good rule of thumb is to prepare to add one more box for system tests of a smaller size, but the same operating system as the target environment. Again, this will be a relatively volatile environment, so it should not be viewed as a place to do industrial-strength testing by a large team.

Integration or Regression Tests
To perform integration and regression tests, it is advisable to have a separate environment that is similar to the target environment. Generally, one server will be enough at this point. However, the contents of this server should be strictly controlled. Either the test coordinator or his or her designate should make all software changes to this environment. Stability and auditability are essential to ensuring the accuracy of test results. Plan for at least one more servers at this stage in testing.

Stress Tests
Stress tests should be done in an environment identical to the target hosting environment. In the first development cycle, this can be done in the production site, before the cutover to production. For subsequent development cycles, a separate environment will have to be maintained for stress testing.
Plan to replicate the deployment environment as part of the test bed for at least the second development cycle of the site. It has also been observed that the most common problem after performance in a high stress environment is database deadlock due to improper programming. Deadlocks are typically difficult to detect and fix and may not show up until the site is highly stressed in production. So it is important that these conditions be stringently tested during the stress test phase.

Acceptance Tests and Staging
Acceptance tests are generally performed in the same environment as the stress tests, so additional hardware is not needed to support this phase of testing. Again, during the initial development cycle, the production environment can be used to perform both acceptance testing and staging, but a new environment should be created for subsequent development cycles.

6. Appendices

7. References

In this subsection:

(1) Provide a complete list of all documents referenced elsewhere in the SRS
(2) Identify each document by title, report number (if applicable), date, and publishing organization
(3) Specify the sources from which the references can be obtained.

This information can be provided by reference to an appendix or to another document. If your application uses specific protocols or RFC's, then reference them here so designers know where to find them.